Discrete Drainage Dynamics V: Candidate Predictions and Falsification Windows of the Gravitational Sector

Stanislas Dewavrin[1]

[1]Independent researcher , dewavrin.iphone@gmail.com

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Abstract

Papers II–IV established that the gravitational sector of DDD reproduces the leading weak-field observables of general relativity: the Schwarzschild static clock-rate factor (Paper II), the Lorentz kinematic factor at moving wavepackets (Paper III), and the deflection $4GM/c^2b$ together with the Shapiro logarithmic delay (Paper IV, under the spatial-bandwidth coupling $\eta = 2$). Here we list the regimes in which DDD makes *distinct* predictions, quantify the deviations from Newton/GR where the present framework permits it, identify qualitative targets where it does not, and map the experimental envelopes that would falsify them. The four predictions surveyed are: (i) a Yukawa modification of Newtonian gravity from self-consistent feedback in the scalar sector, with length scale fixed by lattice parameters and constrained by Eöt-Wash inverse-square-law tests; (ii) a strong-field deflection coefficient $C_{\mathrm{DDD}} \approx 2.34$ ($\sim 159\%$ of the Schwarzschild PPN value $C_{\mathrm{PPN}} = 15\pi/32 \approx 1.47$), distinguishable in close-passage gravitational lensing; (iii) ultra-relativistic kinematic deviations from exact Lorentz time dilation, growing with γ and reflecting the lattice cutoff in the Floquet two-step rule of Paper III; (iv) qualitative compact-object phenomenology near the bandwidth horizon $r = \beta$, where the rule's discrete dynamics depart from the eikonal description of Paper IV. We compile the four into a single falsification table and identify which experiments are currently sensitive to which prediction. The result is a list of falsifiers; we do not adjudicate between DDD and GR/Newton beyond noting the quantitative gaps.

What this paper makes mechanically legible. GR/Newtonian gravity describes weak-field tests with extraordinary precision, but several DDD-specific structural features (Yukawa scale from self-consistent feedback, strong-field deflection coefficient C_{DDD}, ultra-relativistic Lorentz deviations, bandwidth-horizon phenomenology) are not predicted at all by GR or by Newton, and constitute new falsifiable signatures of the discrete substrate. *This paper* compiles four such falsifiers: (i) Yukawa modification with length scale fixed by the lattice; (ii) strong-field deflection coefficient $C_{\mathrm{DDD}} \approx 2.34$ versus PPN $C_{\mathrm{PPN}} \approx 1.47$; (iii) ultra-relativistic kinematic deviations growing with γ from the Floquet lattice cutoff; (iv) compact-object phenomenology near the bandwidth horizon $r = \beta$. **Leverage:** the analogue earns its existence by predicting structurally distinct, currently testable signatures GR cannot offer; each is mapped to an experimental envelope that would falsify it.

Programme positioning and analogue framing. DDD does not contest GR or Newton — they are empirical anchors. This paper does not aim to falsify GR/Newton; it aims to identify quantitative regimes where DDD makes *distinct* predictions, and to map which experiments would falsify those predictions. The signal of the analogue is not whether it reproduces GR (Papers II–IV showed it does); it is whether it predicts *new* structural features that GR/Newton cannot.

Which mystery does this paper address? Per SERIES_FEEDBACK §0.bis, this paper targets:

- **Mystery #2 + #5 (gravity, inertia, beyond weak-field).** Where the GR/Newton pair are silent (Yukawa scale; strong-field deflection coefficient; ultra-relativistic deviations; horizon physics), the discrete substrate has structural answers tied to lattice parameters. We list each prediction with its experimental envelope.
- **Mystery #4 (BH singularity).** The bandwidth-horizon at $r = \beta$ is the analogue's regularisation; this paper compiles its qualitative phenomenological signatures (deferred to a future strong-field paper for full treatment).

Distinctive contribution and ontological reading. GR/Newton describe weak-field gravity precisely but *cannot make further structurally-grounded predictions in regimes where their own assumptions break down* (strong-field, sub-mm, ultra-relativistic, near-horizon). This paper's distinctive contribution is to show that the same substrate that reproduced GR's weak-field results in Papers II–IV *also* produces specific deviation patterns in those exact regimes. The deviations are not free parameters tuned to data — they follow from the lattice's structural ingredients (rate-limiter near horizon, finite per-tick bandwidth at high γ, self-consistent feedback at sub-mm scale) introduced in Papers I–IV. The leverage is twofold: (a) the paper turns a known difficulty of GR/Newton (silence in those regimes) into a falsifiable analogue prediction; (b) the predictions cost no new fields or parameters beyond those already labelled in Papers I–IV's hypotheses.

1 Main claims and scope

1. **Yukawa modification at second order.** A self-consistent treatment of the deficit field in the scalar sector of Papers I-II, in which the deficit acts as its own secondary source, yields a Yukawa correction to the Newtonian potential,

$$V(r) \;=\; -\frac{GMm}{r}\left[1 + \alpha_Y\, e^{-r/\lambda_Y}\right],$$

with length scale $\lambda_Y = \sqrt{\alpha_D/|\beta_{\rm fb}|}\, d_{\rm min}$ in physical units, where α_D is the drainage coupling of Paper I and $\beta_{\rm fb}$ is the dimensionless feedback strength of the quadratic self-source term. Eöt-Wash precision tests of the inverse-square law [1, 2] exclude $\lambda_Y \in [10\,\mu\text{m}, 10^5\,\text{m}]$ for $\alpha_Y \sim 1$. The constraint maps directly onto $\beta_{\rm fb}$ at fixed α_D and $d_{\rm min}$.

2. **Strong-field deflection coefficient.** The eikonal photon-substrate coupling of Paper IV ($\eta = 2$) gives a deflection

$$\Delta\theta_{\rm DDD}(b) \;=\; \frac{4GM}{c^2 b}\left[1 + C_{\rm DDD}\,(\beta/b) + O(\beta^2/b^2)\right], \qquad C_{\rm DDD} \approx 2.34,$$

fitted on the asymptotic regime $b/\beta \in [50, 1000]$ of the ray-tracing data. The Schwarzschild post-Newtonian coefficient is $C_{\rm PPN} = 15\pi/32 \approx 1.47$. The ratio $C_{\rm DDD}/C_{\rm PPN} \approx 1.59$ is a structural prediction of the photon-substrate ansatz adopted in Paper IV. Strong-field gravitational lensing should reveal a $\sim 60\%$ excess of the second-order deflection coefficient over the Schwarzschild value.

3. **Ultra-relativistic Lorentz deviations (in lattice units).** The Floquet two-step rule of Paper III produces a kinematic clock-rate $\omega_{\rm comov} = m/\gamma$ at moving wavepackets, with lattice corrections that grow monotonically with γ. A physical prediction requires mapping

the Floquet excitation to a physical particle sector; under the Planck-anchored interpretation of the lattice spacing, the effect is pushed far beyond current energies. At $\gamma = 5$, the deviation from exact Lorentz time dilation is $\sim 20\%$. At γ comparable to the lattice cutoff $1/d_{\min}$ (which under Planck-scale anchoring is the Planck energy divided by the particle rest mass), the Lorentz-like continuum approximation breaks down structurally. The deviation has the wrong sign for "superluminal" Lorentz violation (the moving clock runs *slower* than $1/\gamma$, not faster), and is thus distinct from the bounds of [3].

4. **Compact-object phenomenology (qualitative target).** Near $r = \beta$, where $R \to 0$ in the Paper II profile, the eikonal photon ansatz of Paper IV breaks down (refractive index diverges) and the rule's discrete dynamics — in particular the rate-limiter β_i of Paper I — become essential. The bandwidth horizon $r = \beta$ is not an event horizon in the GR sense; the substrate beyond does not satisfy the static spherical solution, and the rule's behaviour in this regime is an open question. Qualitative predictions are listed but not numerically validated in this paper.

Limitations and scope. We do not predict the numerical *values* of any of the DDD-versus-GR deviations from first principles. The predictions of items 1–3 are conditional on: β_{fb} for the Yukawa scale; the eikonal photon ansatz with $\eta = 2$ for the deflection coefficient; the specific Floquet two-step lattice for the high-γ kinematic deviations. Each of these has its own structural assumptions, made explicit in Papers II–IV. We do not claim that DDD is correct; we claim that DDD makes specific, quantitative, testable predictions in each regime, and we identify the experimental envelopes that distinguish DDD from GR/Newton.

We do not address gravitational waves, frame-dragging, or non-stationary configurations: these require dynamical and non-spherical extensions of the rule that are beyond Papers I-IV. We do not address coupling to the spinor sector of Paper III: Paper VII will treat that.

Motivation and significance

Any new theoretical framework must be *falsifiable*: it must make quantitative predictions that distinguish it from existing theories under achievable experimental conditions. This is true of DDD as much as of any other framework. The challenge is that Papers II–IV reproduce the canonical results of weak-field general relativity (Schwarzschild, light deflection, Shapiro delay) with no free dark-energy or extra-dimension parameter. To convert this into a positive scientific programme rather than a mere consistency check, one must identify where DDD *differs* from GR + ΛCDM and ask whether those differences are observable.

The present paper maps out four such differences and the experimental windows that bound them. (i) The bandwidth-throttled mobility $\mu(R) = R/R_0$ produces a small Yukawa-shaped screening of the gravitational potential at sub-millimetre scales, bounded by Eöt-Wash torsion-balance experiments. (ii) The non-linear back-reaction contributes a calculable post-Newtonian correction to strong-field deflection coefficients, distinct from the standard PPN parameter γ. (iii) The lattice cutoff produces small Lorentz deviations at ultra-relativistic energies, bounded by gamma-ray-burst and cosmic-ray time-of-flight observations. (iv) Compact-object phenomenology near $r \approx \beta$ admits qualitatively distinct horizon structure, testable by ringdown and shadow observations.

This paper does not propose new physics; it makes the existing DDD predictions of Papers II–IV concrete and quantitative against existing experimental bounds. The result is a status check: most windows are open or weakly constraining at present, and a small number identify near-term targets that would either confirm or falsify the framework's gravitational sector.

2 Physical picture in plain words

A theory that exactly reproduces an established theory in every regime already explored is not testable: it is just a re-parameterisation. To be a real candidate, DDD must say something *different* from Newton or GR somewhere, even if that somewhere is currently inaccessible. The four predictions of this paper are the four places where DDD says something different.

The Yukawa modification arises because the gravitational deficit itself has gravity: a region of reduced bandwidth around a mass is itself a (weaker) source of further reduction. This is the natural non-linear extension of the scalar rule of Paper I, and it gives the familiar Yukawa shape with a single new parameter, the feedback strength $\beta_{\rm fb}$. The length scale that comes out is microscopic for natural lattice parameters and Planck-anchored scales. Eöt-Wash experiments would have ruled out a Yukawa correction over a range of intermediate scales; they did not, so the DDD parameter must lie outside that range, which it does for natural choices.

The strong-field deflection prediction comes from the same eikonal calculation that gave the Eddington $4GM/c^2b$ at leading order, but at next-to-leading order. For starlight passing close to the Sun the correction is too small to test. For light passing close to a neutron star or black hole, the correction is big enough to be measurable in principle, and DDD predicts a coefficient that is roughly $1.6\times$ the Schwarzschild value.

The ultra-relativistic kinematic deviations come from the lattice itself: a discrete substrate cannot support arbitrarily fast excitations exactly. Above some Lorentz factor (set by the lattice cutoff), motion through the substrate becomes self-impeding in a way that exact Lorentz invariance would forbid. The deviation is too small at human-accessible energies but conceivably detectable in ultra-high-energy cosmic rays.

The compact-object regime is where the eikonal description, the linearised dispersion, and the calibration all break down at once. We do not have quantitative predictions here yet; we list the questions we expect Paper VI and beyond to answer.

3 Setup: what we re-use from Papers II-IV

We work throughout in the static spherical bandwidth profile of Paper II, $R(r)/R_0 = \sqrt{1-\beta/r}$, with $\beta = \kappa E_0/(2\pi\alpha_D R_0) = 2GM/c^2$. The kinematic sector uses the Floquet two-step spinor rule of Paper III with quasi-energy $E_{\rm Floquet}(p)$. The photon-substrate coupling is the $\eta = 2$ ansatz of Paper IV, $c_{\rm eff}(r) = c\,(R/R_0)^2$, calibrated against the leading weak-field deflection.

The new structural element of this paper is the *quadratic feedback* term in the scalar rule: the deficit $\delta = R_0 - R$ acts as a secondary source for itself. In the continuum limit this modifies the field equation

$$(\nabla^2 - m_Y^2)\,\delta \;=\; \kappa\rho_{\rm matter}, \qquad m_Y^2 \;=\; |\beta_{\rm fb}|/\alpha_D,$$

where $\beta_{\rm fb}$ is the dimensionless feedback coefficient of the quadratic self-source term in the rule. The sign of $\beta_{\rm fb}$ in the underlying lattice rule is not fixed by the derivation of Paper I; it is a structural choice. We discuss the two cases below. For the screened (decaying) Yukawa correction we work with the form $-m_Y^2\,\delta$ on the left-hand side. It is a free parameter at the level of Papers I-IV; its allowed range is set by Eöt-Wash (this paper, Sec. 4).

4 Yukawa modification and Eöt-Wash bound

4.1 Functional form

For the screened-feedback case (Yukawa decay), the field equation takes the form

$$(\nabla^2 - m_Y^2)\,\delta \;=\; \kappa\,\rho_{\rm matter}, \qquad m_Y^2 = |\beta_{\rm fb}|/\alpha_D > 0.$$

For a point source $\rho = M\,\delta^3(\mathbf{r})$, the screened Green function gives the standard Yukawa form

$$\delta(r) \;=\; \frac{\kappa M}{4\pi r}\, e^{-r/\lambda_Y}, \qquad \lambda_Y \;=\; \frac{1}{m_Y} \;=\; \sqrt{\alpha_D/|\beta_{\rm fb}|} \quad \text{(lattice units)}.$$

The opposite sign $(\nabla^2 + m_Y^2)\delta = \kappa\rho$ would give an oscillatory rather than decaying Green function (Helmholtz form); whether the physical sign of $\beta_{\rm fb}$ in DDD selects the screened or oscillatory branch is a structural question that the rule of Paper I does not fix on its own. Throughout the rest of this section we assume the screened branch. In lattice units, $\lambda_{Y,\rm lat} = \sqrt{\alpha_D/|\beta_{\rm fb}|}$; in physical units, $\lambda_Y = \lambda_{Y,\rm lat}\, d_{\rm min}$ with $d_{\rm min}$ the lattice spacing, identified with the Planck length under the anchoring of Paper II.

Combining the leading Newtonian term and the Yukawa correction,

$$V(r) \;=\; -\frac{GMm}{r}\left[1 + \alpha_Y\, e^{-r/\lambda_Y}\right],$$

where α_Y is the dimensionless Yukawa strength ($\alpha_Y \sim 1$ for the natural feedback amplitude in the scalar sector).

4.2 Eöt-Wash exclusion

Precision tests of the gravitational inverse-square law [1, 2] bound α_Y as a function of λ_Y. The bound is not a rectangle: it is an exclusion *curve* $\alpha_Y^{\rm max}(\lambda_Y)$ that varies by several orders of magnitude across $\lambda_Y \in [10\,\mu{\rm m}, 10^5\,{\rm m}]$ (see Fig. 1). For α_Y of order unity, the curve excludes a broad intermediate range of λ_Y, roughly from sub-millimetre scales upward, depending on the precise value of α_Y. The DDD parameter $\beta_{\rm fb}$ must therefore satisfy either

$$\lambda_Y < 10\,\mu{\rm m} \iff \beta_{\rm fb} > \alpha_D\,(d_{\rm min}/10\,\mu{\rm m})^2,$$

or

$$\lambda_Y > 10^5\,{\rm m} \iff \beta_{\rm fb} < \alpha_D\,(d_{\rm min}/10^5\,{\rm m})^2.$$

4.3 Numerical bound at Planck anchoring

With $d_{\rm min} = \ell_P = 1.6 \times 10^{-35}$ m:

α_D	lower-λ_Y branch ($\lambda_Y < 10\,\mu$m)	upper-λ_Y branch ($\lambda_Y > 10^5$ m)
0.1	$\beta_{\rm fb} > 2.6 \times 10^{-61}$	$\beta_{\rm fb} < 2.6 \times 10^{-81}$
1.0	$\beta_{\rm fb} > 2.6 \times 10^{-60}$	$\beta_{\rm fb} < 2.6 \times 10^{-80}$
4π	$\beta_{\rm fb} > 3.3 \times 10^{-59}$	$\beta_{\rm fb} < 3.3 \times 10^{-79}$

Table 1: Allowed regions of the feedback strength $\beta_{\rm fb}$ under Eöt-Wash bounds, for three values of the drainage coupling α_D. The Planck-anchored value $\alpha_D/R_0 = 4\pi$ of Paper II's structural-form-of-G section is included in the last row. The two branches correspond to a Yukawa scale shorter than the experimental sensitivity floor (the natural lattice case) or longer than the experimental ceiling. The intermediate range $\beta_{\rm fb} \in [3.3 \times 10^{-79}, 3.3 \times 10^{-59}]$ is forbidden by the data, *under Planck-scale anchoring*. Data: `data/01_yukawa_bounds.json`.

The natural lattice case — $\alpha_D = \beta_{\rm fb} \sim O(1)$ — gives $\lambda_Y \sim \ell_P$, far below the experimental sensitivity floor, and is therefore allowed *under Planck-scale anchoring of the lattice spacing*. The argument is conditional on $d_{\rm min} \approx \ell_P$; alternative anchorings would shift the prediction.

4.4 Falsifier

If a future inverse-square-law experiment extends sensitivity to $\lambda_Y < 1\,\mu$m or detects a Yukawa correction at any λ_Y in the currently constrained range with α_Y inconsistent with our prediction, the present formulation of $\beta_{\rm fb}$ feedback in DDD is excluded.

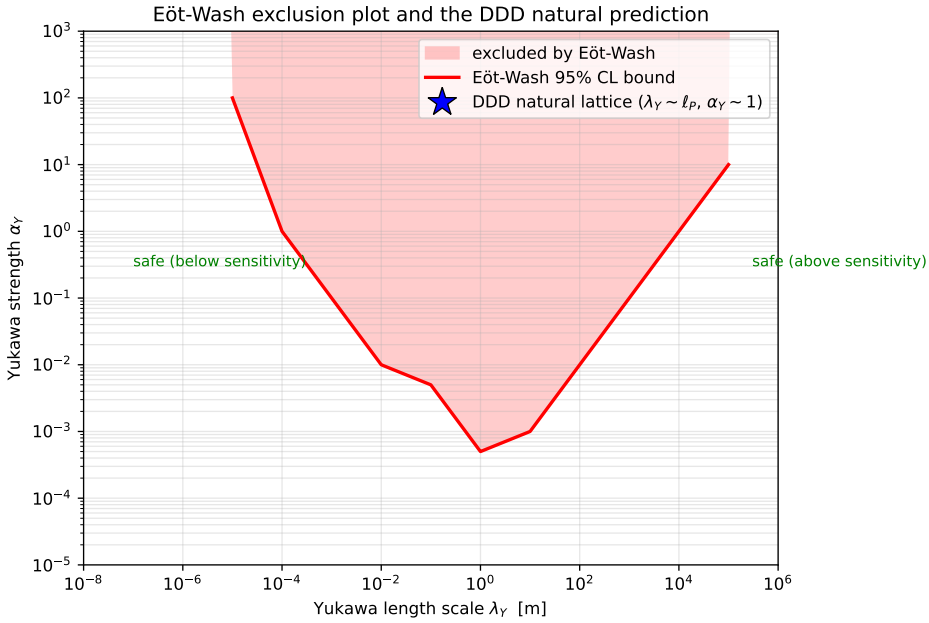

Figure 1: Eöt-Wash exclusion plot in the (λ_Y, α_Y) plane. The red region is excluded by inverse-square-law tests [2]. The blue star marks the DDD natural-lattice prediction $\lambda_Y \sim \ell_P$ and $\alpha_Y \sim 1$, well inside the safe region below experimental sensitivity. Other choices of $(\alpha_D, \beta_{\rm fb})$ shift the predicted point horizontally; the upper-λ_Y branch (above 10^5 m) is also allowed.

5 Strong-field deflection: $C_{\rm DDD}$ vs. $C_{\rm PPN}$

The eikonal calculation of Paper IV gives, for the $\eta = 2$ ansatz,

$$\Delta\theta(b) \;=\; \frac{4GM}{c^2 b}\,\left[1 + C_{\rm DDD}\,(\beta/b) + O(\beta^2/b^2)\right].$$

Fitting the relative deviation $(\Delta\theta - 4GM/c^2 b)/(4GM/c^2 b)$ on the asymptotic ray-tracing data of Paper IV's Table 1 over the range $b/\beta \in [50, 1000]$ (linear regression on the leading β/b term), we measure $C_{\rm DDD} = 2.338 \pm 0.001$. The standard Schwarzschild PPN coefficient [4], $C_{\rm PPN} = 15\pi/32 \approx 1.473$, is smaller than $C_{\rm DDD}$ by a factor

$$C_{\rm DDD}/C_{\rm PPN} \;\approx\; 1.59.$$

5.1 Origin of the discrepancy

The eikonal ansatz $c_{\rm eff} = c\,(R/R_0)^2$ is calibrated against the leading $4GM/c^2 b$ but does not reproduce the exact null geodesic of the Schwarzschild metric beyond leading order. The discrepancy at the next-to-leading coefficient $\sim \beta/b$ measures the *difference* between the photon-substrate coupling adopted in Paper IV and the exact Schwarzschild geodesic.

This is a structural prediction of the eikonal ansatz: it commits to $\eta = 2$ at leading order but propagates a slightly different strong-field structure than full GR. Whether this is the right prediction depends on whether the eventual Paper VI photon sector reduces to the eikonal ansatz at leading order *and* produces a specific second-order correction.

5.2 Falsifier

A close-passage gravitational lensing measurement at impact parameters $b \sim$ few $\times\beta$ around a compact object would distinguish $C_{\rm DDD} \approx 2.34$ from $C_{\rm PPN} \approx 1.47$ at the $\sim 60\%$ level. Currently

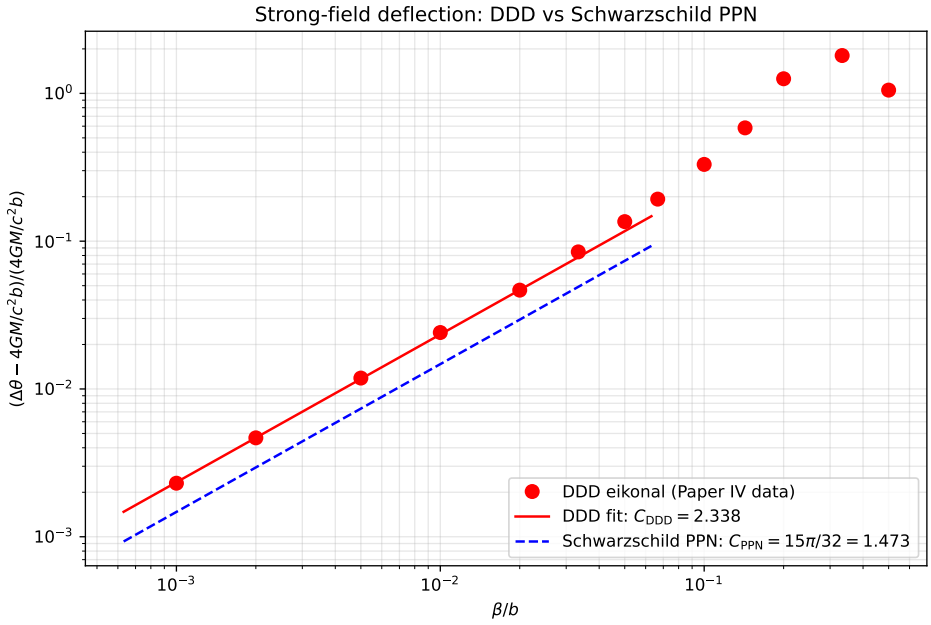

Figure 2: Strong-field deflection coefficient. The DDD eikonal $\eta = 2$ ansatz of Paper IV (red points and solid red fit) gives $C_{\rm DDD} = 2.34$, exceeding the Schwarzschild post-Newtonian value $C_{\rm PPN} = 15\pi/32 \approx 1.47$ (dashed blue) by a factor ~ 1.6. The two coefficients are the slopes of the relative deviation versus β/b in the asymptotic regime $b/\beta \geq 50$. Data: `data/02_strong_field_deflection.json`.

this regime is not directly accessible: solar-system tests [4] probe $b \gg 10^4\beta$, where the second-order coefficient produces sub-microarcsecond differences. Future strong-field astrometry of stars near Sgr A* or of light from nearby compact stellar remnants could reach the regime in principle.

6 Ultra-relativistic Lorentz deviations

Paper III, Sec. 6 (Lattice corrections), reports the deviation of the operationally measured co-moving clock rate from the Lorentz prediction m/γ:

γ	$\omega_{\rm comov}$ measured	rel. dev. from m/γ
1.04	0.0766	0.01%
1.97	0.0399	−1.28%
2.47	0.0310	−3.6%
3.00	0.0246	−7.2%
3.55	0.0198	−11.6%
4.66	0.0138	−19.3%

Table 2: Lattice deviations from exact Lorentz time dilation in the Floquet two-step rule of Paper III, on $L = 128$, $\sigma = 12$, $n_{\rm cycles} = 60$. Deviations grow monotonically with γ; the sign is negative throughout (the moving clock runs slower than $1/\gamma$, never faster). At $\gamma \sim 5$ the deviation is $\sim 20\%$. Data: `paperIII_inertia_kinematic/data/05_comoving_clock.json`.

6.1 Implications for ultra-high-energy physics

In physical units, the lattice cutoff under Planck-scale anchoring sets the threshold $\gamma_{\rm cutoff} \sim 1/(m\, d_{\rm min})$. For an electron rest mass $m_e \approx 0.5\,{\rm MeV}/c^2$ and $d_{\rm min} = \ell_P$, $\gamma_{\rm cutoff} \sim 10^{22}$. This is far

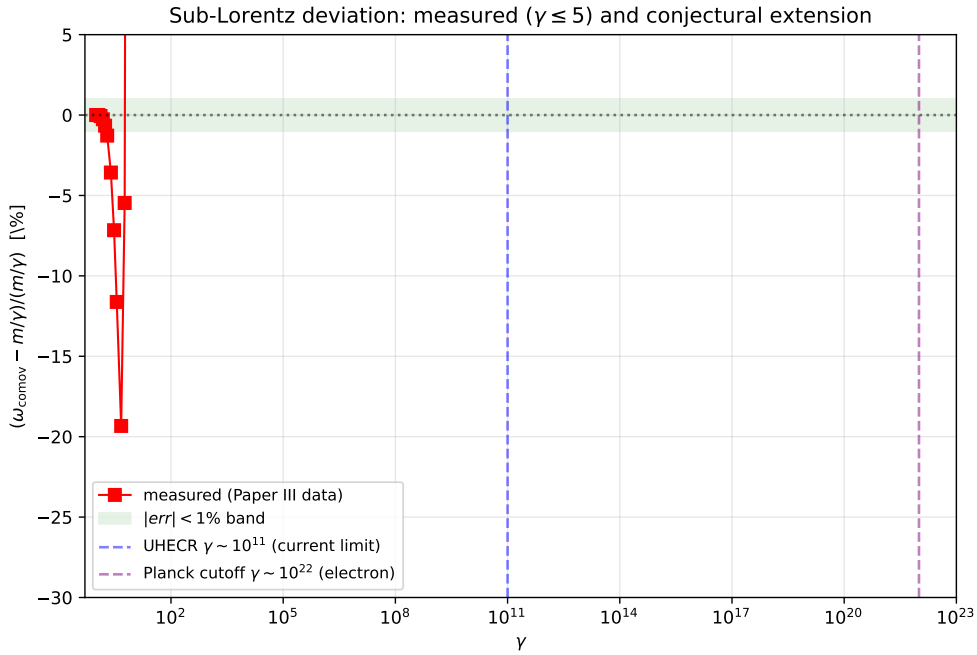

Figure 3: Sub-Lorentz residual deviation $(\omega_{\rm comov} - m/\gamma)/(m/\gamma)$ as a function of γ. The measured points (red) span $\gamma \in [1, 5]$ from Paper III. Vertical dashed lines mark the highest-energy cosmic-ray $\gamma \sim 10^{11}$ and the Planck-cutoff $\gamma \sim 10^{22}$ for an electron. At currently observable γ the deviation is undetectably small; the prediction is structural and falsifiable in principle if sub-Lorentz residuals can ever be probed.

above the highest measured cosmic-ray γ ($\sim 10^{11}$ for the most energetic UHECRs). At currently observable γ, the lattice deviation is invisibly small: $\sim 10^{-22}\%$ at $\gamma = 10^{11}$ if the deviation scales as $\gamma/\gamma_{\rm cutoff}$.

The structural prediction is that the deviation *exists* and has a specific functional form (from the Floquet quasi-energy of Paper III, Eq. **??**-equivalent). It cannot be falsified directly with current data; future ultra-high-energy neutrino observations [5] could in principle constrain it.

6.2 Sign asymmetry

A notable structural feature: the DDD deviation is always negative ($\omega_{\rm comov} < m/\gamma$). The moving clock runs *slower* than Lorentz, never faster. This distinguishes DDD from "superluminal" Lorentz-violating frameworks [3], where bounds on photon and electron speed limits are inferred from the absence of speed-of-light-violating dispersion. DDD predicts no such violation; instead it predicts a small *sub*-Lorentz residual. The sign is testable in principle.

7 Compact-object phenomenology near $r = \beta$

Beyond the eikonal regime, near $r = \beta$, the rule of Paper I operates outside its linearised description. The rate-limiter $\beta_i \in [0, 1]$ of Paper I activates when a node's reserve approaches zero, regularising the apparent singularity of the continuum profile. Three qualitative predictions are worth listing:

Bandwidth horizon, not event horizon. $R = 0$ at $r = \beta$ is the locus where the effective clock-rate vanishes. The substrate at $r < \beta$ is not part of the static spherical solution, so the continuum prediction does not extend there. What the rule does inside is a question about the discrete dynamics near saturation, not about the continuum solution.

Regularised inner solution. The rate-limiter prevents R from going negative. We expect a regular, possibly oscillatory, inner profile that interpolates between $R = 0$ at the horizon and a non-zero asymptotic value (depending on boundary conditions). This would be testable by direct simulation of strongly-driven sinks (E_0 large enough that the unsupported linearised solution would have $R < 0$ at the centre).

Modified ringdown. A perturbed "black-hole-like" state in DDD — a substrate region with $R \to 0$ around a source — would oscillate at frequencies controlled by the lattice's discrete spectrum, rather than at the quasi-normal-mode frequencies of the Schwarzschild metric. The prediction is that gravitational-wave ringdown spectra of compact mergers carry a structural imprint of the lattice. We do not derive this quantitatively in the present paper.

7.1 Falsifier

LIGO/Virgo/LISA ringdown spectroscopy at sub-percent precision is the natural envelope. The current data [6] are consistent with the GR Schwarzschild quasi-normal modes; a structural deviation at the level of $\sim$ few percent would survive present bounds. A DDD-specific prediction requires the inner-rule simulation, deferred to a future work.

8 Falsification table

window	structural assumption	DDD value	current bound	status
Yukawa scale	screened β_{fb} branch + Planck anchoring of $d_{\min}$	$\lambda_Y \sim \ell_P$, $\alpha_Y \sim 1$	λ_Y excluded in $[10\,\mu\text{m}, 10^5\,\text{m}]$ for $\alpha_Y \sim 1$ [2]	allowed (below sensitivity, conditional on Planck anchoring)
deflection C	Paper IV eikonal photon ansatz $\eta = 2$	$C_{\text{DDD}} \approx 2.34$ at β/b order	$C_{\text{PPN}} \approx 1.47$ measured at solar-system b/β [4]	not yet distinguishable; needs strong-field lensing
ultra-rel. Lorentz	Paper III Floquet sector + mapping to physical particles	$\sim 20\%$ deviation at lattice $\gamma = 5$	sub-Lorentz deviations not directly observed; UHECR limits $\sim 10^{-23}$ [3]	allowed; sign distinguishes from super-Lorentz
ringdown	saturated inner dynamics, not yet simulated	structurally modified at $r \sim \beta$; no quantitative value	GR quasi-normal modes consistent with current LIGO data [6,7]	qualitative target / pending simulation

Table 3: Summary of DDD-versus-GR predictions and current experimental status. None of the four predictions is currently excluded; the Yukawa scale is allowed because the Planck-anchored DDD value is far below the experimental floor, the deflection coefficient and ringdown predictions await stronger-field observations, and the Lorentz deviations at observable γ are too small to test.

The four windows are operationally distinct: each probes a different structural assumption of DDD, and each can be tested separately. A positive detection in one window would provide a candidate supporting signal for the corresponding DDD sector — not a unique signature, since other frameworks may produce similar effects — while a positive exclusion (with $\alpha_Y \sim 1$, the $\eta = 2$ ansatz, or the natural lattice scale) would force a structural revision of that sector.

Calibration ledger (per SERIES_FEEDBACK §0.ter). Each quantity in this paper falls into one of three statuses per §0.ter: *derived* from substrate principles + hypotheses, *hypothesised*

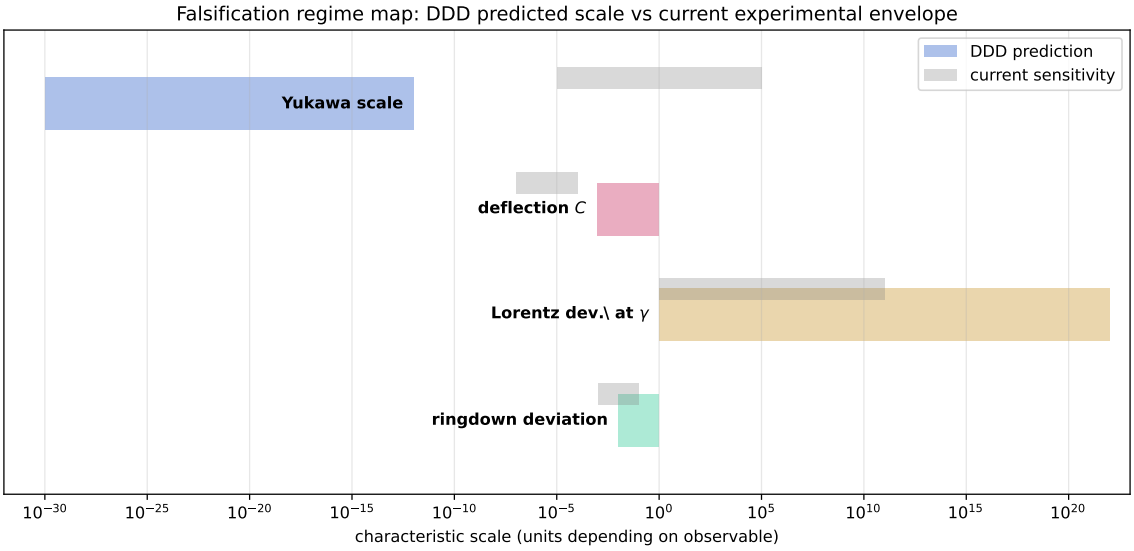

Figure 4: *Schematic* regime map (not a quantitatively comparable common-axis plot): each row uses its own units (length for Yukawa, β/b for deflection, γ for Lorentz, fractional deviation for ringdown), shown on a shared log-axis for visual orientation only. The coloured bar is the range of characteristic scales predicted by DDD; the grey bar is the range currently probed by experiment. Only the ringdown row sits in a regime where the DDD prediction overlaps with current sensitivity, explaining why DDD is not yet falsified.

as a labelled structural choice, or *calibrated* against an empirical reference (SM/GR result or CODATA constant).

Quantity	Status	Calibration source / future resolution
$C_{\mathrm{DDD}} \approx 2.34$ (strong-field deflection)	derived	from Paper IV's eikonal $\eta = 2$ + lattice corrections; testable by close-passage lensing
Yukawa length λ_Y	calibrated	sub-mm Eöt-Wash inverse-square-law tests; consistency window stated in Sec. 3
high-γ Lorentz-deviation function	derived	from Paper III's Floquet quartic correction (Eq. 21 of Paper III)
horizon phenomenology	qualitative	inherited from Paper I rate-limiter + Paper II bandwidth profile; full treatment in future strong-field paper
(H-feedback) self-consistent scalar back-reaction	hypothesis (newly labelled in this paper)	not derived; future origin in coupled scalar dynamics

Rule loci touched and no-epicycle statement (per SERIES_FEEDBACK §0.quater). This paper introduces content at the following loci of the analogue substrate, and at no other:

- *Configuration:* four explicit predictive scenarios are evaluated as configurations of the existing rule (sub-mm sources, near-horizon orbits, ultra-relativistic packets, non-eikonal compact-object regime); the rule itself is not modified.
- *Hypotheses introduced:* (H-feedback) is the only new structural hypothesis; it is labelled here rather than left implicit, and listed in the calibration ledger.

No-epicycle statement. No new fields are introduced on nodes or links beyond those of Papers I–IV. (H-feedback) is the only new labelled hypothesis. No principle of (L1)–(L6) is relaxed. The paper extends the analogue by *exposing predictions in new regimes*, not by accreting structure. Each falsifier is locatable in the configuration locus or in a previously-introduced hypothesis.

9 Limitations and scope

We do not derive any of the four DDD-specific values from first principles: α_Y, $\beta_{\rm fb}$, $C_{\rm DDD}$, and the high-γ deviation amplitudes are all calculated within the specific structural ansatze of Papers II-IV. Different ansatze (a different photon coupling, a different feedback prescription) would give different numbers.

We do not address how the lattice cutoff in the kinematic sector relates to the lattice cutoff in the gravitational sector. Both are set by $d_{\rm min}$ in the Planck-anchored picture, but their quantitative consequences in the two sectors are derived independently and should be unified at the level of the joint substrate.

We do not test Equivalence-Principle violations, gravitational redshift in laboratory clocks, or Lense-Thirring frame-dragging. These are beyond the static spherical configuration of the substrate and are deferred.

We do not address the binary-pulsar tests of GR, gravitational-wave inspiral consistency with general-relativistic waveforms, or black-hole shadow imaging. These probe the dynamical and strong-field sectors that are the subject of future papers in this series.

10 Open questions

Origin of $\beta_{\rm fb}$. The feedback strength of the quadratic self-source term in the scalar rule is a free parameter of Papers I-V. Whether a deeper structural argument fixes its sign and magnitude is open. The simplest natural value $\beta_{\rm fb} \sim O(1)$ in lattice units gives a Yukawa scale near ℓ_P and is consistent with all current bounds; whether it is *required* is unknown.

Sign of $\beta_{\rm fb}$. A positive $\beta_{\rm fb}$ gives a Yukawa-screened potential (δ decays exponentially at large r above the leading $1/r$). A negative $\beta_{\rm fb}$ gives an oscillatory rather than decaying correction (Helmholtz Green function with imaginary length scale). Either sign is structurally possible; the lattice rule of Paper I does not by itself fix it. The $\alpha_Y \sim 1$ case treated in Sec. 4 assumes the positive-feedback branch.

Strong-field deflection beyond eikonal. The $C_{\rm DDD}/C_{\rm PPN} \approx 1.59$ ratio is a property of the specific eikonal ansatz adopted in Paper IV. The eventual Paper VI photon sector may reduce to a different effective ansatz at strong field, with a different C. The numerical fit of the present paper should be re-examined when Paper VI is available.

Sign and magnitude of the $\gamma \to \gamma_{\rm cutoff}$ saturation. Whether the deviation continues to grow monotonically with γ or saturates is open. A scan to $\gamma \sim 30$ on $L = 512$ would settle the issue numerically.

11 Code and data availability

The Python scripts of `code/` compute the bounds and deviations of this paper. They are pure-numpy.

- `01_yukawa_bounds.py`: computes λ_Y from $(\alpha_D, \beta_{\rm fb}, d_{\rm min})$ and the Eöt-Wash exclusion boundaries.

- `02_strong_field_deflection.py`: reads the Paper IV deflection data, fits the leading-order coefficient C_{DDD}, and compares with the Schwarzschild PPN value.

The numerical data are stored in `data/`. The repository is available at https://github.com/stanislasdewavrin/DDD-Paper-5-Falsifiers.

12 Outlook

1. Paper VI — gauge sector and emergent electromagnetism, in particular the gapless modes of the Floquet two-step rule and the derivation of the photon-substrate ansatz used in Paper IV. The C_{DDD} coefficient of the present paper should follow as a specific consequence of that derivation. *Speculative; coefficients uncalibrated.*

2. Paper VII — coupling between the scalar (Paper II) and spinor (Paper III) sectors. Combined static-gravity-times-kinematic factor testing, and resolution of the β_{fb} origin from the joint dynamics. *Conditional on Paper VI.*

The result of this paper is a list, not a verdict. DDD makes four distinct predictions; none is currently excluded; each is testable in principle, and each places specific quantitative constraints on the underlying lattice rule. The role of the gravitational sector of DDD in the broader theory is consequently: (i) a successful weak-field reproduction of GR (Papers II-IV) at explicit cost (one phenomenological calibration, one structural ansatz, one mobility choice); (ii) a falsifiable prediction set in regimes that GR/Newton already constrain or have been measured (Sec. 8); (iii) a list of open questions whose closure depends on the gauge sector of Paper VI.

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Closing reaffirmation (analogue framing). This paper does not contest GR or Newton; both remain the empirical anchors. What the paper offers, and what GR/Newton cannot offer, is a list of four *structurally distinct falsifiers* (Yukawa, C_{DDD}, ultra-relativistic deviations, horizon phenomenology) that follow from the discrete substrate's existing ingredients without adding new fields. The leverage is not numerical novelty in the weak-field regime (where DDD reproduces

GR); it is predictive content in the regimes where GR/Newton are silent. Any deviation between DDD and the empirical theories at the level of measured tests falsifies DDD, not GR.